# Machine Learning Project 1
# Predicting Student Exam Scores Using Regression and Feature Selection

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**Abstract**

This project investigates the prediction of student exam scores using regression-based machine learning models and feature selection techniques. The dataset contains academic, behavioral, and lifestyle-related variables such as study hours, attendance, sleep hours, exercise frequency, social media usage, diet quality, internet quality, gender, and mental health rating. The objective is to identify the most influential factors affecting exam performance and to build accurate and interpretable predictive models.

The analysis includes exploratory data analysis, missing value assessment, correlation analysis, hypothesis testing, baseline regression modeling, regularized regression, and multiple feature selection strategies. Three baseline models were evaluated: Linear Regression, Ridge Regression, and Lasso Regression. In addition, four feature selection methods were applied: Forward Selection, Backward Elimination, Hybrid Selection, and Best Subset Selection.

The results show that study hours per day is the strongest predictor of exam score, with a strong positive Pearson correlation of 0.8254. Among the baseline models, Lasso Regression performed best, achieving a test $R^2$ of 0.8982. However, the overall best-performing model was Best Subset Selection, which achieved an $R^2$ of 0.8996 and an RMSE of 5.0745 using only 7 features. These findings indicate that student exam performance can be predicted accurately using a compact set of behavioral and lifestyle-related variables.

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# 1 Introduction

Predicting student academic performance is an important problem in educational data analysis because it helps identify the factors that most strongly influence learning outcomes. In many real-world settings, student performance depends not only on study behavior, but also on lifestyle habits, mental health, attendance, and access to resources. Understanding these relationships can support better academic planning and intervention strategies.

The aim of this project is to predict student exam scores using supervised machine learning methods, with a particular focus on regression models. In addition to building predictive models, the project investigates the relative importance of different variables and evaluates several statistical hypotheses related to academic performance.

The main objectives of this project are:

- To perform exploratory data analysis and understand the structure of the dataset.
- To identify statistically significant relationships between selected variables and exam score.
- To train and compare several regression models, including Linear, Ridge, and Lasso Regression.
- To apply feature selection methods to improve interpretability and predictive performance.
- To select the final model based on predictive accuracy and simplicity.

# 2 Dataset Description

The dataset contains student-related attributes that can plausibly influence exam performance. These variables include both numerical and categorical features covering academic behavior, digital media usage, health, and lifestyle.

Examples of important variables include:

- `study_hours_per_day`
- `attendance_percentage`
- `sleep_hours`
- `exercise_frequency`
- `social_media_hours`
- `netflix_hours`
- `mental_health_rating`
- `diet_quality`
- `internet_quality`
- `gender`

The target variable is:

- `exam_score`

The project treats exam score as a continuous variable, making regression the appropriate predictive modeling framework.

# 3 Exploratory Data Analysis

Exploratory data analysis was conducted to understand the distribution of the target variable, assess missing values, and inspect relationships among predictors.

## 3.1 Missing Values

The dataset contained a limited number of missing values across several columns. These missing values were handled during preprocessing to ensure that the machine learning pipeline could be trained consistently.

Table 1: Missing values summary

| Column Name | Missing Count | Missing Percentage |
|---|---|---|
| teacher_quality | 78 | 1.18% |
| parental_education_level | 91 | 9.10% |
| distance_from_home | 67 | 1.02% |

The proportion of missing values was relatively small, so standard preprocessing methods were sufficient to handle them without major information loss.

## 3.2 Distribution of Exam Scores

The target variable, exam score, was visualized to understand its distribution.

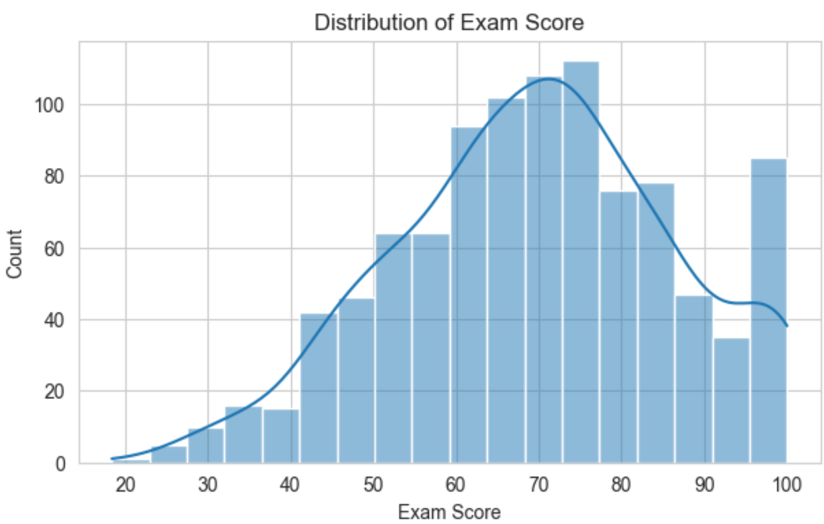


Figure 1: Distribution of exam scores

The distribution appears reasonably continuous and suitable for regression modeling. No extreme irregularities were observed that would invalidate the use of linear regression-based models.

### 3.3 Correlation Analysis

A correlation matrix was used to examine the relationships among numerical variables and the target variable.

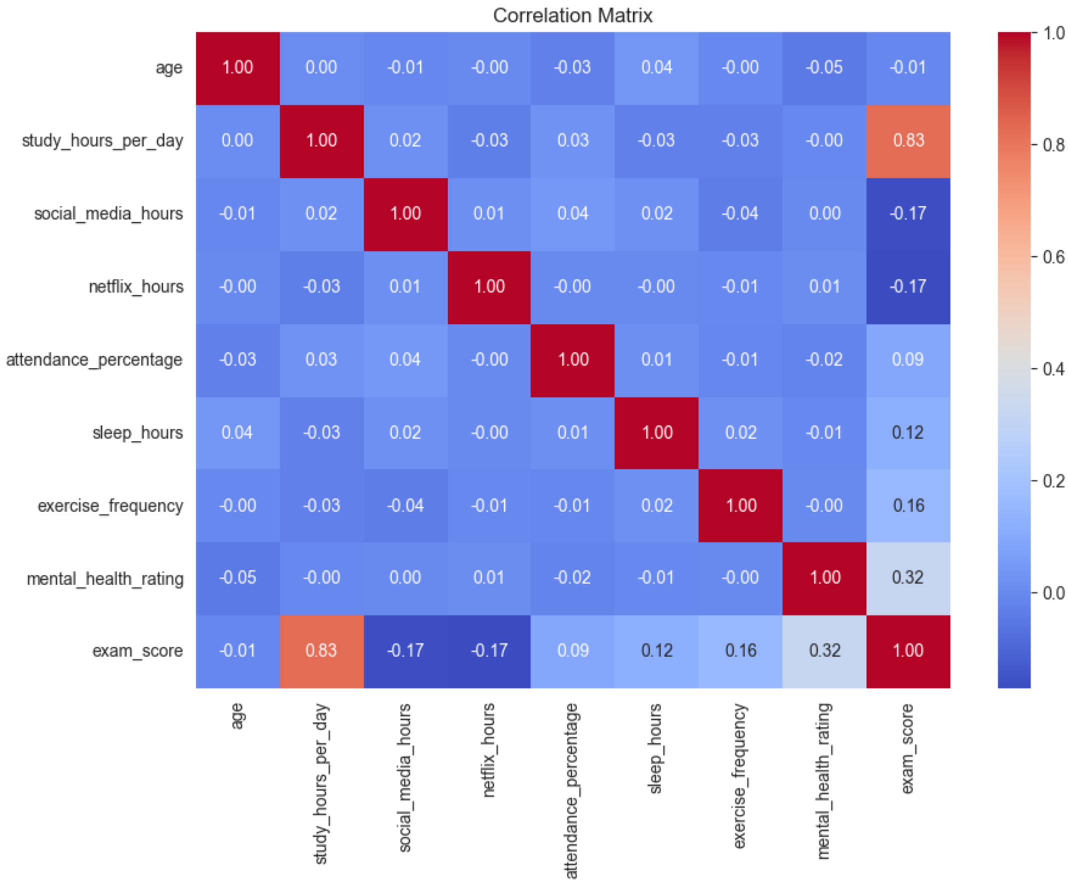


Figure 2: Correlation matrix of numerical variables

The correlation matrix shows that `study_hours_per_day` has the strongest positive relationship with `exam_score`. Several other variables, including attendance percentage, sleep hours, exercise frequency, and mental health rating, also show meaningful relationships with exam performance. In contrast, variables such as social media hours and Netflix hours tend to show weaker or negative relationships.

Overall, the exploratory analysis suggests that academic behavior and lifestyle-related numerical variables are likely to be the most useful predictors.

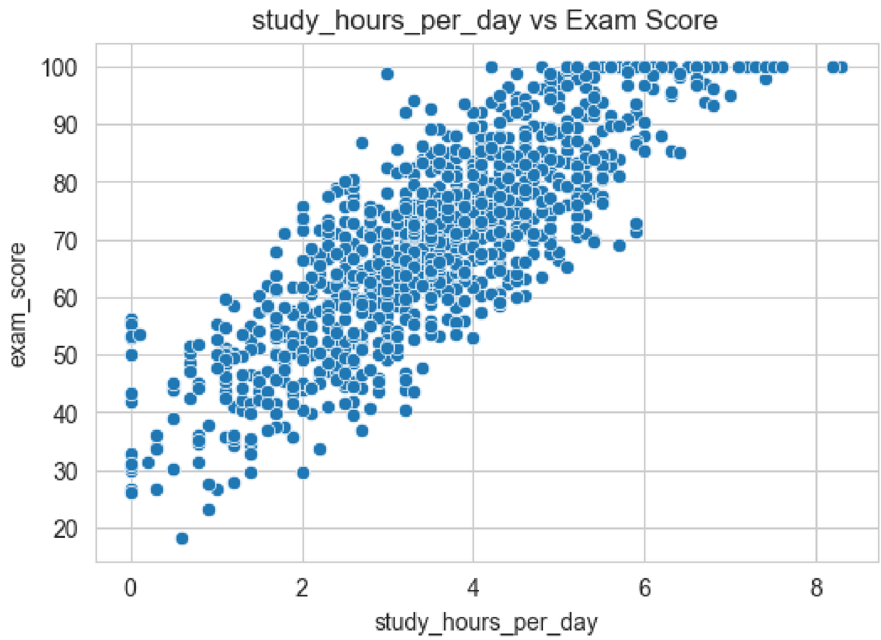


Figure 3: Relationship between study hours per day and exam score

Figure 3 visually confirms the strong positive relationship between study hours per day and exam score. Students who spend more time studying per day generally achieve higher exam scores, which is consistent with the Pearson correlation result.

## 4 Preprocessing and Data Leakage Prevention

Proper preprocessing is essential for producing reliable machine learning results. In this project, preprocessing was implemented using a pipeline-based workflow to ensure that all transformations were fitted only on the training data and then applied to the test data. This design prevents data leakage.

Numerical and categorical variables were processed separately using a `ColumnTransformer`. Numerical features were scaled, while categorical features were encoded using one-hot encoding. Missing values were also handled within the preprocessing framework.

The use of a `Pipeline` provided two major benefits:

- It ensured that preprocessing steps were applied consistently across training, validation, and test sets.
- It prevented information from the test set from influencing model training or feature engineering decisions.

This approach is especially important when using regularized models and feature selection methods, because leakage can artificially inflate performance metrics.

## 5 Hypothesis Testing

Several statistical hypotheses were tested to evaluate whether selected factors are significantly associated with exam score. A significance level of $\alpha = 0.05$ was used for all hypothesis tests.

Table 2: Summary of hypothesis testing results

| Hypothesis | Test Type | Statistic | $p$-value | Conclusion |
|---|---|---|---|---|
| Study hours and exam score | Pearson correlation | 0.8254 | 0.0000 | Reject $H_0$; statistically significant positive relationship |
| Part-time job and exam score | Independent samples t-test | -0.8523 | 0.3946 | Fail to reject $H_0$; not statistically significant |
| Attendance and exam score | t-test | 2.6865 | 0.0073 | Reject $H_0$; statistically significant difference |
| Sleep hours and exam score | Spearman correlation | 0.1234 | 0.0001 | Reject $H_0$; statistically significant relationship |
| Diet quality and exam score | ANOVA | 1.2662 | 0.2824 | Fail to reject $H_0$; not statistically significant |
| Gender and exam score | ANOVA | 0.1423 | 0.8674 | Fail to reject $H_0$; not statistically significant |
| Internet quality and exam score | Kruskal-Wallis | 3.2213 | 0.1998 | Fail to reject $H_0$; not statistically significant |
| Extracurricular participation and high exam score | Chi-square test | 0.0238 | 0.8774 | Fail to reject $H_0$; not statistically significant |

### 5.1 Hypothesis: Study Hours and Exam Score

This hypothesis examines whether study hours per day are linearly associated with exam score.

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

A Pearson correlation test was performed and produced the following result:

$$r = 0.8254$$

$$p\text{-value} = 0.0000$$

Since the $p$-value is less than 0.05, the null hypothesis is rejected. Therefore, there is a strong and statistically significant positive relationship between study hours per day and exam score.

### 5.2 Hypothesis: Attendance and Exam Score

This hypothesis investigates whether students with higher attendance have significantly different exam scores compared with students with lower attendance.

$$H_0 : \mu_{\text{high attendance}} = \mu_{\text{low attendance}}$$

$$H_1 : \mu_{\text{high attendance}} \neq \mu_{\text{low attendance}}$$

A t-test was conducted to compare the exam scores of the high-attendance and low-attendance groups. The test produced the following result:

$$t = 2.6865$$

$$p\text{-value} = 0.0073$$

Since the $p$-value is less than 0.05, the null hypothesis is rejected. Therefore, there is statistically significant evidence that exam scores differ between students with high attendance and students with low attendance.

### 5.3 Hypothesis: Sleep Hours and Exam Score

This hypothesis examines whether sleep hours are associated with exam performance.

$$H_0 : \rho_s = 0$$

$$H_1 : \rho_s \neq 0$$

A Spearman correlation test was used and produced the following result:

$$\rho_s = 0.1234$$

$$p\text{-value} = 0.0001$$

Because the $p$-value is less than 0.05, the null hypothesis is rejected. This indicates a statistically significant relationship between sleep hours and exam score. However, the relationship is weak in practical magnitude.

### 5.4 Hypothesis: Part-time Job and Exam Score

This hypothesis examines whether students with a part-time job have significantly different exam scores compared with students without a part-time job.

$$H_0 : \mu_{\text{part-time job}} = \mu_{\text{no part-time job}}$$

$$H_1 : \mu_{\text{part-time job}} \neq \mu_{\text{no part-time job}}$$

An independent samples t-test was conducted to compare the two groups. The result was:

$$t = -0.8523$$

$$p\text{-value} = 0.3946$$

Since the $p$-value is greater than 0.05, the null hypothesis is not rejected. Therefore, there is no statistically significant evidence that students with and without part-time jobs differ in their exam scores.

### 5.5 Hypothesis: Diet Quality and Exam Score

This hypothesis investigates whether exam scores differ across different diet quality groups.

$$H_0 : \mu_{\text{Poor}} = \mu_{\text{Fair}} = \mu_{\text{Good}}$$

$$H_1 : \text{At least one group mean differs}$$

A one-way ANOVA test was used and produced the following result:

$$F = 1.2662$$

$$p\text{-value} = 0.2824$$

Because the $p$-value is greater than 0.05, the null hypothesis is not rejected. Therefore, there is no statistically significant evidence that mean exam scores differ across diet quality groups.

### 5.6 Hypothesis: Gender and Exam Score

This hypothesis examines whether mean exam scores differ across gender groups. Since the gender variable contains three categories, including an "Other" category, a one-way ANOVA test was used instead of an independent samples t-test.

$$H_0 : \mu_{\text{Male}} = \mu_{\text{Female}} = \mu_{\text{Other}}$$

$$H_1 : \text{At least one gender group has a different mean exam score}$$

A one-way ANOVA test was conducted to compare exam scores across the gender categories. The result was:

$$F = 0.1423$$

$$p\text{-value} = 0.8674$$

Since the $p$-value is greater than 0.05, the null hypothesis is not rejected. Therefore, there is no statistically significant evidence that mean exam scores differ across gender groups.

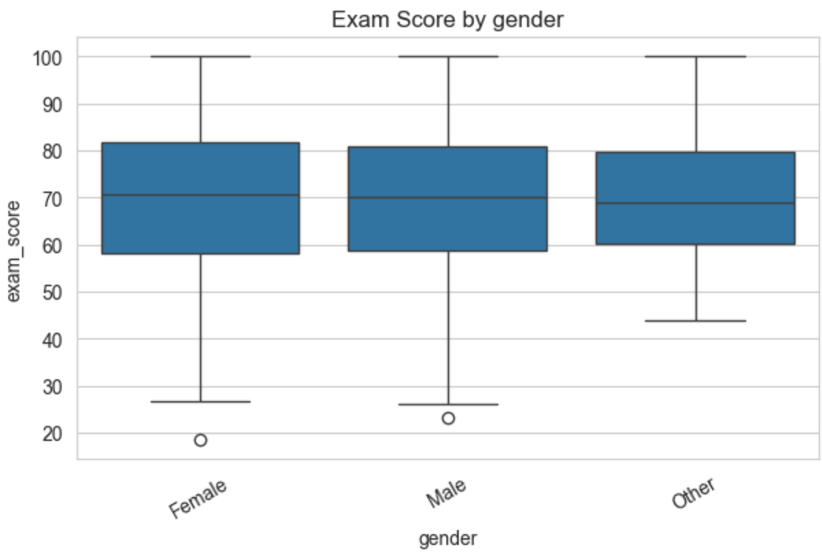


Figure 4: Distribution of exam scores across gender categories

Figure 4 shows that the exam score distributions across gender groups are broadly similar. This supports the ANOVA result, which indicated no statistically significant difference in mean exam scores among gender categories.

### 5.7 Hypothesis: Internet Quality and Exam Score

This hypothesis investigates whether exam scores differ across different internet quality groups. Since internet quality is a categorical variable and the assumptions of parametric ANOVA may not always be appropriate, the Kruskal-Wallis test was used.

$$H_0 : \text{The distributions of exam scores are the same across internet quality groups}$$

$$H_1 : \text{At least one internet quality group has a different exam score distribution}$$

A Kruskal-Wallis test was conducted and produced the following result:

$$H = 3.2213$$

$$p\text{-value} = 0.1998$$

Since the $p$-value is greater than 0.05, the null hypothesis is not rejected. Therefore, there is no statistically significant evidence that exam score distributions differ across internet quality groups.

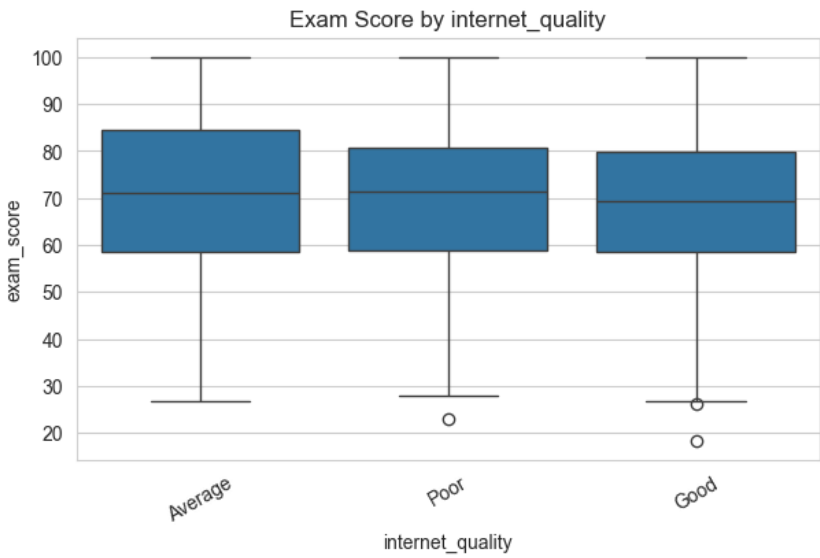


Figure 5: Distribution of exam scores across internet quality groups

Figure 5 shows the distribution of exam scores across different internet quality groups. The distributions appear relatively similar, which is consistent with the Kruskal-Wallis test result showing no statistically significant difference among the groups.

### 5.8 Hypothesis: Extracurricular Participation and High Exam Score

This hypothesis tests whether extracurricular participation is associated with achieving a high exam score.

$$H_0 : \text{Extracurricular participation and high exam score are independent}$$

$$H_1 : \text{Extracurricular participation and high exam score are not independent}$$

A chi-square test of independence was conducted. The result was:

$$\chi^2 = 0.0238$$

$$p\text{-value} = 0.8774$$

Since the $p$-value is greater than 0.05, the null hypothesis is not rejected. Therefore, there is no statistically significant evidence of an association between extracurricular participation and achieving a high exam score.

## 6 Regression Models

Three baseline regression models were evaluated: Linear Regression, Ridge Regression, and Lasso Regression.

## 6.1 Linear Regression

Ordinary Least Squares Linear Regression was used as the first baseline model.

The Linear Regression model achieved the following results:

$$R^2 = 0.8968$$

$$\text{Adjusted } R^2 = 0.8889$$

$$MAE = 4.1893$$

$$RMSE = 5.1455$$

The 5-fold cross-validation result was:

$$\text{CV } R^2 = 0.8955 \pm 0.0106$$

These results indicate that the linear baseline model explains approximately 89.7% of the variance in exam scores and generalizes well.

## 6.2 Ridge Regression

Ridge Regression was applied as a regularized linear regression model using an $L_2$ penalty. The regularization parameter $\alpha$ was selected using cross-validation.

The best selected value of $\alpha$ was:

$$\alpha = 10.0$$

The Ridge Regression model achieved the following results:

$$R^2 = 0.8964$$

$$\text{Adjusted } R^2 = 0.8885$$

$$MAE = 4.1923$$

$$RMSE = 5.1545$$

The 5-fold cross-validation result was:

$$\text{CV } R^2 = 0.8958 \pm 0.0101$$

Compared with Linear Regression, Ridge Regression produced very similar performance. The selected alpha value of 10.0 indicates that moderate regularization was preferred.

### 6.3 Lasso Regression

Lasso Regression was used as a regularized linear model with an $L_1$ penalty. Unlike Ridge Regression, Lasso can shrink some coefficients exactly to zero and therefore can act as an embedded feature selection method.

The best selected regularization parameter was:

$$\alpha = 0.1$$

The Lasso Regression model achieved the following results:

$$R^2 = 0.8982$$

$$\text{Adjusted } R^2 = 0.8905$$

$$MAE = 4.1506$$

$$RMSE = 5.1096$$

The 5-fold cross-validation result was:

$$\text{CV } R^2 = 0.8970 \pm 0.0104$$

Among the three baseline regression models, Lasso achieved the best performance. Its predictive accuracy was slightly better than both Linear Regression and Ridge Regression.

## 7 Bias-Variance Tradeoff

The bias-variance tradeoff is an important concept in machine learning model selection. A model with high bias is too simple and may underfit the data, while a model with high variance is too sensitive to the training data and may overfit.

Linear Regression is relatively simple and can perform well when the relationship between predictors and the target is approximately linear. Ridge and Lasso Regression introduce regularization to reduce model variance by shrinking coefficients. Ridge does this smoothly using an $L_2$ penalty, while Lasso uses an $L_1$ penalty that can also eliminate less useful features.

In this project, the three baseline models showed very similar performance, which suggests that the underlying signal in the data is strong and relatively stable. The small advantage of Lasso indicates that mild regularization and embedded feature selection can slightly improve generalization.

## 8 Feature Selection

Feature selection methods were applied to identify a smaller subset of predictors that could maintain or improve predictive performance.

## 8.1 Feature Selection Results

Four feature selection strategies were evaluated: Forward Selection, Backward Elimination, Hybrid Selection, and Best Subset Selection.

Forward Selection selected 9 features and achieved:

$$R^2 = 0.8992$$

$$\text{Adjusted } R^2 = 0.8944$$

$$MAE = 4.1273$$

$$RMSE = 5.0853$$

$$\text{CV } R^2 = 0.8976$$

Backward Elimination and Hybrid Selection achieved identical performance and selected the same number of features.

Best Subset Selection achieved the strongest overall result while using only 7 features:

$$R^2 = 0.8996$$

$$\text{Adjusted } R^2 = 0.8959$$

$$MAE = 4.1139$$

$$RMSE = 5.0745$$

$$\text{CV } R^2 = 0.8975$$

Overall, the feature selection methods slightly improved predictive performance while reducing model complexity. Best Subset Selection was the best-performing method.

## 8.2 Selected Features

Table 3: Selected features by different feature selection methods

| Method | Selected Features |
|---|---|
| Forward Selection | study_hours_per_day, mental_health_rating, social_media_hours, exercise_frequency, sleep_hours, netflix_hours, attendance_percentage, diet_quality_Good, internet_quality_Good |
| Backward Elimination | study_hours_per_day, social_media_hours, netflix_hours, attendance_percentage, sleep_hours, exercise_frequency, mental_health_rating, diet_quality_Good, internet_quality_Good |
| Hybrid Selection | study_hours_per_day, mental_health_rating, social_media_hours, exercise_frequency, sleep_hours, netflix_hours, attendance_percentage, diet_quality_Good, internet_quality_Good |
| Best Subset Selection | study_hours_per_day, attendance_percentage, sleep_hours, social_media_hours, netflix_hours, mental_health_rating, exercise_frequency |

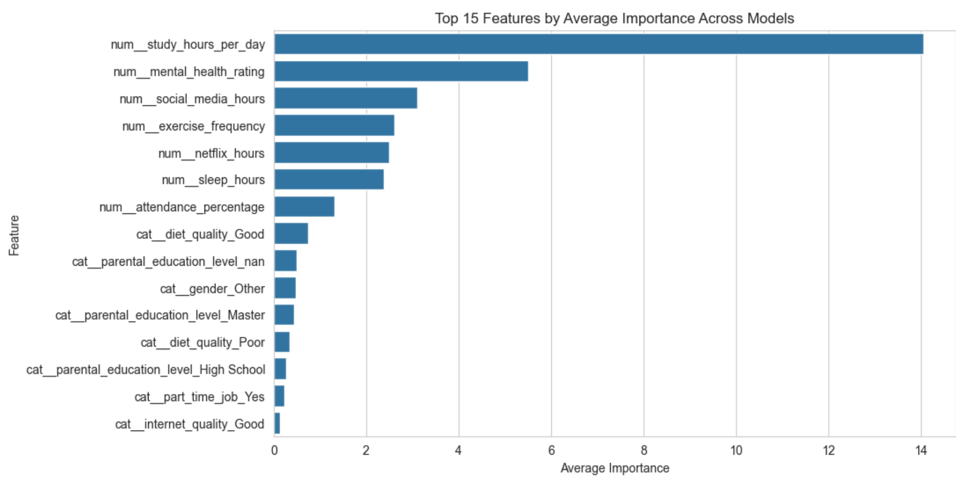


Figure 6: Average feature importance across selection or modeling procedures

Figure 6 shows the average importance of the predictors across the feature selection or modeling procedures. The figure confirms that study hours per day is the most influential feature in predicting exam score. Other important predictors include attendance percentage, sleep hours, exercise frequency, mental health rating, and media usage variables. This result is consistent with the correlation analysis, hypothesis testing, and the final regression models.

## 9 Model Comparison

Table 4: Comparison of baseline regression models and feature selection approaches

| Approach | Category | Best $\alpha$ | $R^2$ | Adjusted $R^2$ | MAE | RMSE | CV $R^2$ | CV $R^2$ Std | Num. Features |
|---|---|---|---|---|---|---|---|---|---|
| Best Subset Selection | Feature Selection | – | 0.8996 | 0.8959 | 4.1139 | 5.0745 | 0.8975 | – | 7 |
| Forward Selection | Feature Selection | – | 0.8992 | 0.8944 | 4.1273 | 5.0853 | 0.8976 | – | 9 |
| Hybrid Selection | Feature Selection | – | 0.8992 | 0.8944 | 4.1273 | 5.0853 | 0.8976 | – | 9 |
| Backward Elimination | Feature Selection | – | 0.8992 | 0.8944 | 4.1273 | 5.0853 | 0.8976 | – | 9 |
| Lasso Regression | Baseline Model | 0.1 | 0.8982 | 0.8905 | 4.1506 | 5.1096 | 0.8970 | 0.0104 | 14 |
| Linear Regression | Baseline Model | – | 0.8968 | 0.8889 | 4.1893 | 5.1455 | 0.8955 | 0.0106 | 14 |
| Ridge Regression | Baseline Model | 10.0 | 0.8964 | 0.8885 | 4.1923 | 5.1545 | 0.8958 | 0.0101 | 14 |

The results show that all approaches performed well, with $R^2$ values around 0.896–0.900. Among the baseline models, Lasso Regression performed best. However, the feature selection methods slightly improved performance further. Best Subset Selection achieved the highest $R^2$, the lowest MAE, and the lowest RMSE while using the fewest features.

## 10 Discussion

The analysis shows that student exam performance is strongly influenced by a relatively small group of behavioral and lifestyle-related variables. Study hours per day was consistently the most important predictor and showed the strongest statistical relationship with exam score. This result is both intuitive and strongly supported by the correlation analysis and regression results.

Attendance percentage, sleep hours, exercise frequency, mental health rating, social media usage, and Netflix usage also contributed to predictive performance. These findings suggest that student success is not determined by academic effort alone, but is also connected to broader daily habits and well-being.

The hypothesis testing results showed that study hours, attendance, and sleep hours were statistically significant. In contrast, part-time job status, diet quality, gender, internet quality, and extracurricular participation were not statistically significant based on their respective tests. This suggests that, in this dataset, the most meaningful predictors are mainly related to direct academic behavior and daily habits rather than demographic or broad categorical variables.

Interestingly, Best Subset Selection excluded categorical features and still achieved the best overall performance. This suggests that the key predictive information in this dataset is concentrated in numerical variables. The fact that a 7-feature model performed better than the full baseline models also indicates that removing less informative variables can improve interpretability without sacrificing accuracy.

## 11 Final Model Selection

Based on the evaluation results, Best Subset Selection was chosen as the final recommended model. It achieved the best overall performance among all evaluated approaches:

$$R^2 = 0.8996$$

$$\text{Adjusted } R^2 = 0.8959$$

$$MAE = 4.1139$$

$$RMSE = 5.0745$$

The model used only 7 selected features:

$$\{\texttt{study\_hours\_per\_day}, \texttt{attendance\_percentage}, \texttt{sleep\_hours}, \texttt{social\_media\_hours}, \texttt{netflix\_hours}, \texttt{mental\_}$$

Compared with the baseline models, Best Subset Selection achieved slightly better predictive accuracy while using fewer features. Therefore, it provides the best balance between predictive performance and interpretability.

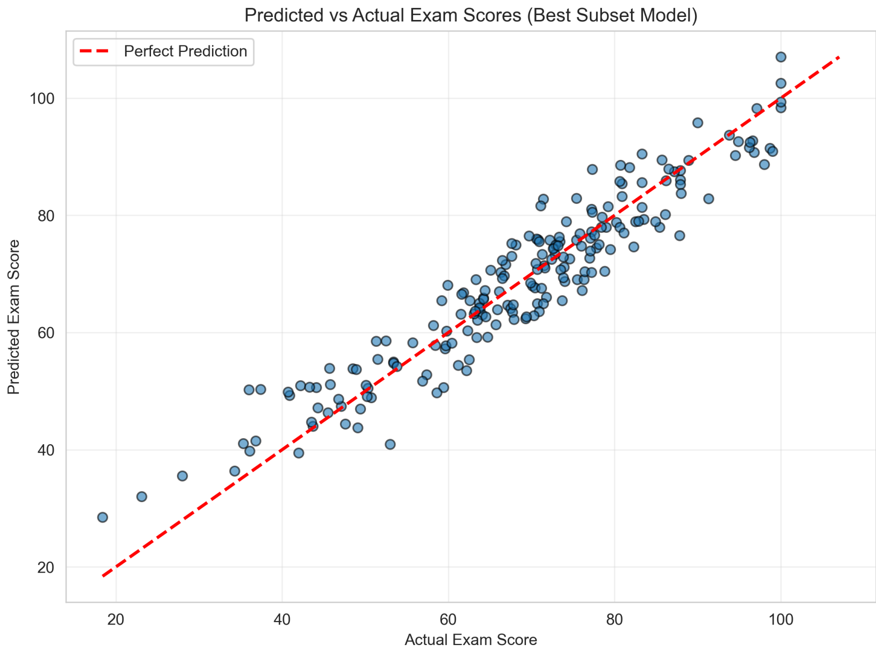


Figure 7: Predicted versus actual exam scores for the final selected model

Figure 7 compares the actual exam scores with the predicted scores from the final selected model. The points are generally close to the diagonal reference line, indicating that the model provides accurate predictions and captures the main patterns in the data.

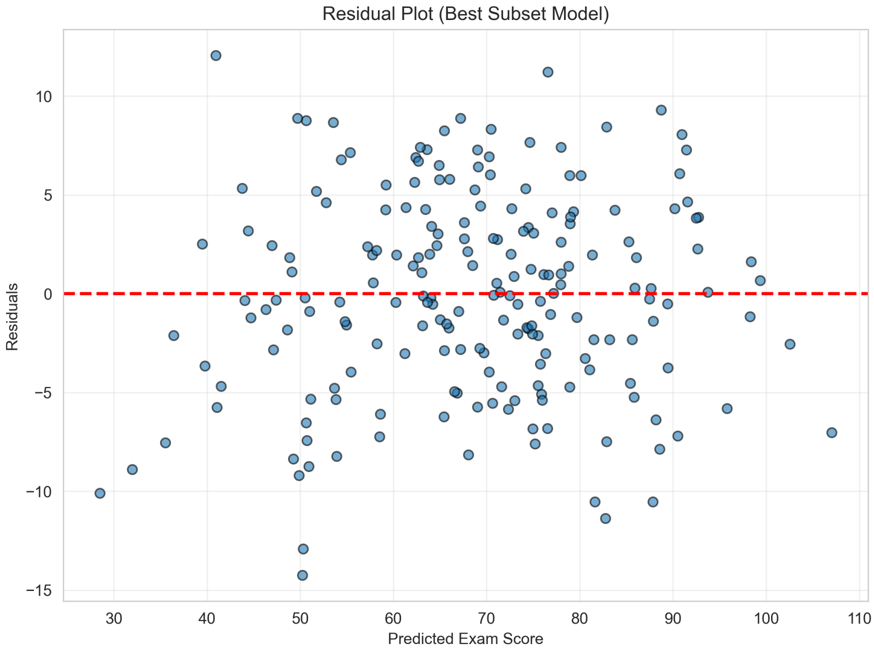


Figure 8: Residual plot for the final selected model

Figure 8 shows the residuals of the final selected model against the predicted exam scores. The residuals are centered around zero, suggesting that the model does not show a strong systematic prediction bias.

## 12 Conclusion

This project aimed to predict student exam scores using regression-based machine learning models and feature selection techniques. The results showed that exam performance can be predicted accurately using a compact set of behavioral, academic, and lifestyle-related variables.

Among the baseline models, Lasso Regression achieved the best performance, with an $R^2$ of 0.8982 and an RMSE of 5.1096. Ridge Regression selected an optimal regularization parameter of $\alpha = 10.0$, while Lasso Regression selected $\alpha = 0.1$. Although all baseline models performed similarly, Lasso showed a slight advantage.

Feature selection methods further improved performance. Best Subset Selection achieved the best overall result, with an $R^2$ of 0.8996, an adjusted $R^2$ of 0.8959, an MAE of 4.1139, and an RMSE of 5.0745 using only 7 features. The selected features were study hours per day, attendance percentage, sleep hours, social media hours, Netflix hours, mental health rating, and exercise frequency.

The hypothesis testing results supported several important findings. Study hours had a strong and statistically significant positive relationship with exam score. Attendance and sleep hours were also statistically significant. In contrast, part-time job status, diet quality, gender, internet quality, and extracurricular participation were not statistically significant in the corresponding tests.

Overall, the final recommended model is the Best Subset Selection model because it provides the best balance between predictive accuracy and interpretability. The results suggest that student exam performance is mainly influenced by study habits, attendance, mental health, sleep, exercise, and digital media usage.